he Specification: Tech ingredients

A specification reads like a list of project features, describing the unit, and will usually include:

- Inputs
- Controls
- Outputs
- Indicators
- Functions
- Modes of operation
- Power Supply
- Communications
- Protection, Fail safes and replaceable parts
- Connector types
- Physical format and size

The Big List of Bullet points

A list of technical terms to help you (we hope!) with your specification

Inputs

- Keyboard and Switch inputs
    - A Keyboard is usually used in association with a display, for instance LCD, LED,
    - At the concept stage, the exact function of every key is often not critical - it can be set in software later
    - Keyswitch types
        - Existing keyboards - e.g. IBM PC keyboards: reliable, replaceable, cheap, reasonably easy to interface
        - Membrane keyboards - custom manufactured flexible plastic laminate keyboards
        - Flexible rubber keyboards - shaped raised keys, need large production volumes to be viable
        - Individual Tactile or pushbutton keys - A keyboard made up of individual switches in a custom arrangement
            - 6 x 6mm tactile keys
            - 12 x 12mm tactile keys
            - Several other types, including lit pushbuttons
    - DIP switches - PCB mounted switches used for setting configuration options
    - Jumpers - not strictly a switch, but used as one - three posts with a moveable two pin conducting sleeve
    - Slide switches - usually used for on/off functions or mode/option setting
    - Thumbwheel switches - switches with an up/down button or wheel to dial a number with digits 0-9 or 0-F
    - Rotary switches - older fashioned panel mount switches with a knob that turns in steps

    - Lock switches - barrel type switches operated with a key, to provide secured access to an operation
- Other switch and relay contact inputs
    - Microswitches
      Tough level/cam switches used to measure mechanical position or as limit switches
    - Relay contacts
      Used for isolation of circuits, more especially in industrial plant environments
- Opto-isolated inputs
  Used to isolate the input circuit from the main electronics
- Opto-interrupters
  Used to measure position - a slotted disk or a metal tab interrupts an infra-red light beam to send the signal
- Analog inputs - voltage and current
  Characterised by the number of bits of resolution
    - 8 bits - 256 steps of measurement
    - 10 bits - 1024 steps of measurement
    - 12 bits - 4096 steps of measurement
    - 16 bits - 65536 steps of measurement
- Potentiometers
  Rotary controls, resistance varies according to angle of shaft rotation
- Temperature sensors
    - Silicon temperature sensors - medium accuracy 0-100 degree sensors, often easily interfaced
    - Diode or transistor - The cheapest, nastiest, temperature sensor, usually very cheap to implement
    - Thermocouples - accurate, wide range temperature measurement requiring low level front end electronics
    - Thermistor types - readily available, cheaper temperature measurement devices requiring calibration, linearisation
    - Platinum resistance sensors (PTD's) - highly accurate sensors, require linearisation but generally no calibration
- Light Sensors
    - Photodiodes - accurate, fast response sensors, need some front end circuitry
    - Phototransistors - basically amplified photodiodes, usually very easy to apply for basic light sensing
    - LDR - light dependent resistor
    - OPIC - optical IC, a whole range of optical sensors with integrated electronics are available for specialist applications
- Magnetic Field sensors
    - Hall effect devices - generally small 3 pin devices activated by a strong magnet within close proximity
    - Reed switches - simple 2 pin switches activated by a strong magnet within close proximity
- Strain gauges
  Used for measuring weight, by strain of a support. Configured as a bridge, with sophisticated front end circuitry.

Controls

Controls are often an abstraction of inputs - a potentiometer input may be used to *control the speed of a motor*, or a keyboard switch may be used to *ramp the* *motor speed down to zero.*

Outputs

- Parallel outputs, TTL level
  - Standard type of output from single chip microprocessors
  - "TTL level" has become nomenclature for a type of interface, rather than the use of old fashioned TTL logic
  - TTL level signals may be protected against extraneous voltages through the use of clamp diode circuitry
  - Typical sink ratings vary from a "MOS drive" level of 1.6ma, to "CMOS" at 6ma, to "Buffered Outputs" at 24ma
  - TTL level signals are the typical levels used to drive LEDs and optocouplers
- Open collector outputs
  - Open collector outputs sink current to ground, but in the "hi" state are essentially open circuit
  - "Open Collector" has become nomenclature for an output topology rather than the use of TTL/Bipolar devices
  - Open collector outputs allow reliable connection between two units with separate, independent power supplies
- High current outputs
  - Typically implemented with Transistor driver arrays such as the octal device ULN2803A or UDN2981
  - High Current outputs are more robust and better able to handle extraneous voltages than TTL level outputs
  - Most High current outputs are also open collector types
  - Typical ratings are 100mA - 800mA per output, with a maximum rating for the driver package as a whole
  - High current outputs are typically used to drive relays
- Analog outputs
  - Typically implemented using a Digital-to-Analog (D-to-A or D/A) converter, but sometimes Pulse Width Modulation (PWM)
  - Analog outputs provide a variable output voltage or current
- Relays
  - Relay outputs provide electrical isolation and are generally very robust
  - Relay outputs allow reasonably heavy loads to be driven: 0.5A - 15A for PCB mount devices
  - In most Industrial applications, the PCB mount relay will be used to drive an external relay or contactor that then runs the load to be switched.
- Solid State Relays (SSRs)
  - SSR's are invariably optically coupled, and usually switch 110VAC/240VAC
  - SSRs use triac/optocoupler/snubber circuitry integrated into one encapsulated package
- Optically isolated outputs
  Optically isolated outputs use optocouplers to provide electrical isolation
- Servos
  - Servos are usually driven using TTL level signals with a 20ms period pulse width modulation
  - Servos require driver firmware, but are otherwise easily interfaced to single chip microprocessors

- Solenoids
  - Solenoids provide quick stroke axial motion
  - Solenoids are usually used as a mechanical actuating mechanism - e.g. Solenoid controlled valves for liquids

Indicators

- Light Emitting Diodes (LED's)
  - Colour: Red, Green, Yellow, Orange (only marginally different from Red/Yellow), Blue (expensive)
  - Size: 3mm and 5mm are standard, miniature surface mount, 2mm, 8mm, 10mm available
  - Diffuse: Available waterclear, colour tinted and diffuse, with correspondingly wider viewing angle
  - Intensity: Available in standard, high efficiency/low current, high brightness, super bright
  - Shape: Standard dome, also available in rectangular, special types: flat top, square, arrowhead
  - Circuitry: Some special types with integral current limiting or flashing circuits are available
  - Full Colour types: LEDs with Red-Green-Blue elements are now available, if somewhat expensive
- LED displays - LED bar
  - Arrays of Rectangular LEDs encapsulated in DIP package block
  - Standard size is 10 indicators in one bar, can be stacked for longer lengths of display
  - Normally Red, but available in other colours, even mixed in one package (Green--Orange--Red)
- Lamps - Incandescent
  - Available in sizes from 2mm up
  - Brightness is inversely proportional to operating life
  - High current requirements
  - Standard Yellowish-White Available with coloured filter caps, including colour adjust to bright white
- Audible Indicators
  - Piezo sounders/speakers - output volume low to medium, may be driven to make many types of sound
  - Piezo buzzers - fixed frequency, output volume medium to unbearably high
  - Speakers - conventional magnetic speakers are available in many shapes, including substantially flat

Indicators: Displays

- LED displays - 7 segment with decimal point
  - Standard Red and Green
  - Standard sizes 0.3 inch, 0.5 inch, 0.8 inch
  - While highly visible for indoor applications, LEDs are hopeless in sunlight
- LED displays - alphanumeric, 14/16 segment
- LED displays - 5x7 matrix
- Liquid Crystal Displays (LCD) - 1,2 or 4 lines of characters

    - Standard type: 2 lines by 12 characters to 4 lines of 40 characters (Typical: 2 lines by 16 characters)
    - Available with backlight
    - Available in extended temperature range
    - Onboard controller - display is accessed by parallel bus and appropriate firmware drivers
- LCD: Displays - Graphic type
    - Typically higher cost than the standard character modules
    - Available in 64 x 64 pixels to 256 x 128 pixels as standard
    - Available with controller (more easily interfaced) or without (cheaper, more costly interfacing)
    - Available with backlight (However LED backlights often consume excessive amounts of power)
- LCD: Displays - colour Graphic type
    - Available in volume
    - If a product needs a colour LCD Graphic display and keyboard, the product might more economically be manufactured by writing suitable software for a laptop PC

Functions

Describe functions simply and generally. Describe special cases separately

Modes of operation

Modes of operation affect functions, for instance *Powering up, manual operation, automatic operation*

Power supply

Power supplies to be connected to mains need approval according to country:

- US - Underwriters Laboratories (UL) approval
- Canada - Canadian Standards Association (CSA)
- Australia - Individual state energy authority approval
- New Zealand - Compliance to NZ or Australian equivalent standard
- UK - British Standards approval (BS)

- Plug Pack, or wall adaptor - reasonably priced for power up to 10-15W
    - DC Type - standard. Unregulated, standard nominal voltages are 6v, 9v, 12v, currents 150ma-1Amp
    - AC Type - used for high wattages, transformer fills plugpack body. Standard 12v, 16v, 1-1.5 Amp
    - DC Regulated Type - available but not standard.
- Switchmode, Internal with IEC connector - similar to PC power supply, reasonably priced for 15-65W
- Battery
    - Non-rechargeable
        - Silver oxide (Watch batteries) - very small, suitable only for low power devices
        - Carbon Zinc, Standard batteries
        - Alkaline, Higher capacity replacement for standard batteries

- Rechargeable
  - Nickel Cadmium - standard rechargeables, suffer from memory effect
  - High Temperature Nickel Cadmium
  - Nickel Metal Hydride - improved standard rechargeables, less memory effect, poor standby performance
  - Lead Acid battery - car battery type
  - Gel Cell - Lead Acid battery with gelled electrolyte, much less leakage, available in smaller sizes

Communications

Describe the medium (e.g. *RS232*) the data (e.g. *Aluminium thickness measurements*) and the protocol.